Retrieval-Augmented Summarization of Scientific Articles

1 Overview

This study evaluates multiple retrieval-augmented generation (RAG) strategies for generating summaries of scientific papers from the arXiv Summarization Dataset. The system combines dense retrieval, passage reranking, and large language model generation to produce summaries grounded in retrieved document passages.

Several RAG variants were investigated, including different chunking strategies, retrieval depths, reranking configurations, and passage summarization strategies. The experiments examine:

1) retrieval configuration and chunking strategy;
2) faithfulness metrics (sentence grounding, entailment, coverage);
3) reference-based summarization metrics (ROUGE and BERTScore);
4) runtime efficiency and retrieval depth trade-offs.

The results demonstrate how retrieval design choices influence summary grounding, semantic similarity to reference abstracts, and computational efficiency, highlighting trade-offs between retrieval precision, generation quality, and runtime cost.

2 Model Overview

The RAG pipeline combines three core components:

1) dense semantic retrieval using the SentenceTransformer model all-MiniLM-L6-v2 to encode document chunks and queries;
2) passage reranking using the cross-encoder model, which refines the ranking of retrieved passages based on query–passage relevance;
3) text generation using Phi-2 language model to generate summaries conditioned on retrieved evidence.

Faithfulness of generated summaries was evaluated using an entailment model based on RoBERTa-large-MNLI. The following evaluation metrics are computed allowing evaluation of both reference similarity and faithfulness to retrieved evidence:

- sentence grounding (average semantic similarity between generated sentences and retrieved evidence passages);
- entailment score (logical support of generated sentences by retrieved passages using a natural language inference model);
- coverage (fraction of generated sentences whose semantic similarity to retrieved evidence exceeds a predefined threshold);
- ROUGE (lexical similarity between generated summaries and reference abstracts);
- BERTScore (semantic similarity between generated summaries and reference abstracts).

3 Variant Performance Comparison

3.1 Overall Performance

To compare configurations, a variant-level composite score was computed by averaging normalized values of sentence grounding, ROUGE-1, and coverage. This composite metric captures both faithfulness to retrieved evidence (grounding and coverage) and similarity to reference summaries (ROUGE-1). The highest performing configuration that significantly outperformed the baseline configuration was:

Variant	Chunk Size	Overlap	Composite Score
small_chunks_high_overlap	100	75	**0.748**

Key observations

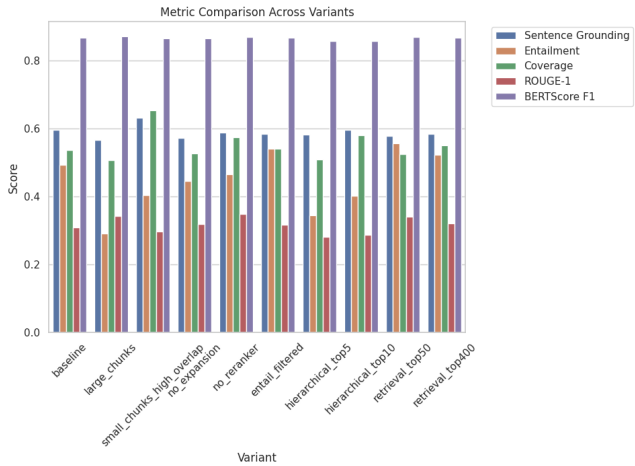
Smaller chunks with high overlap improve performance because they:

- produce more focused semantic passages;

- increase the probability that relevant evidence is retrieved;
- reduce the chance of missing information near chunk boundaries.

As a result, generated summaries show higher grounding and higher coverage, indicating stronger alignment between generated sentences and retrieved context.

As illustrated in the Metric Comparison Across Variants figure, the small_chunks_high_overlap configuration achieves the highest sentence grounding and coverage scores among all variants.



3.2 Metric Comparison

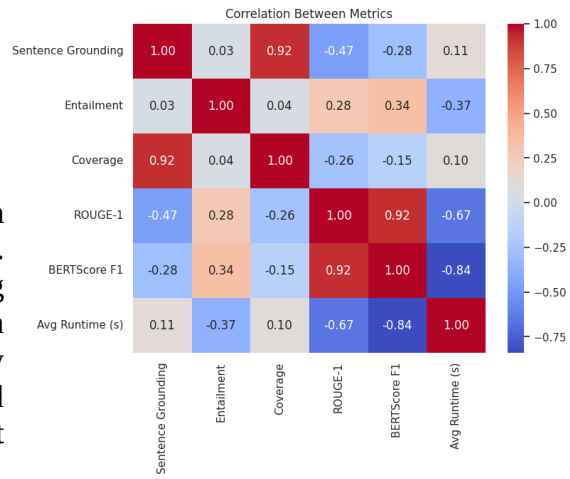
The metric comparison across variants shows several patterns:

- **sentence grounding** (the highest grounding score is achieved by the small_chunks_high_overlap configuration: sentence grounding ≈ 0.631. This indicates that sentences generated by the model are more semantically aligned with retrieved passages when smaller overlapping chunks are used);
- **entailment** (highest entailment scores occur for retrieval_top50: 0.557 and entail_filtered: 0.541. These variants produce summaries that are more frequently logically supported by retrieved evidence according to the NLI model);
- **coverage** (the highest coverage is also achieved by small_chunks_high_overlap with coverage ≈ 0.653. This indicates that most generated sentences have supporting passages in the retrieved context).

3.3 ROUGE and Semantic Similarity

Reference-based metrics are summarized below:

Variant	ROUGE-1	BERTScore F1
no_reranker	**0.348**	0.869
large_chunks	0.343	**0.871**



These results suggest that removing reranking can slightly increase similarity to reference abstracts. However, this does not necessarily improve grounding or coverage. As shown in the Correlation Between Metrics figure, ROUGE-1 and BERTScore are strongly correlated, while their relationship with grounding and coverage is weaker. This reflects an important distinction:

- ROUGE and BERTScore measure similarity to reference summaries;
- grounding and coverage measure factual support from retrieved evidence.

Therefore, a configuration can achieve strong reference similarity while still being weakly grounded in retrieved context.

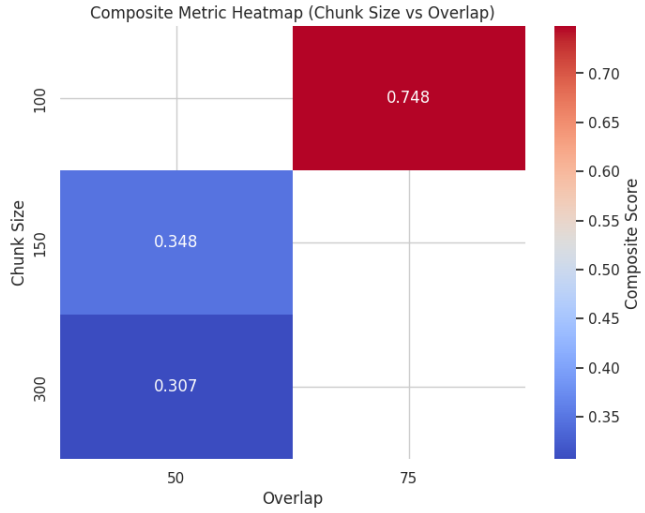
4 Retrieval and Chunking Effects

4.1 Chunk Size and Overlap

The impact of chunk configuration is visualized in Figure Composite Metric Heatmap. Best configuration was:

Chunk Size	Overlap	Composite Score
100	75	0.748

In this implementation, chunk sizes are measured in words, not tokens. Smaller chunks improve retrieval precision by isolating semantically coherent text segments, while higher overlap prevents information loss at chunk boundaries. In contrast, larger chunks (300 words) reduce retrieval precision because each chunk contains more heterogeneous information, making it harder for the retriever to identify the most relevant passages.

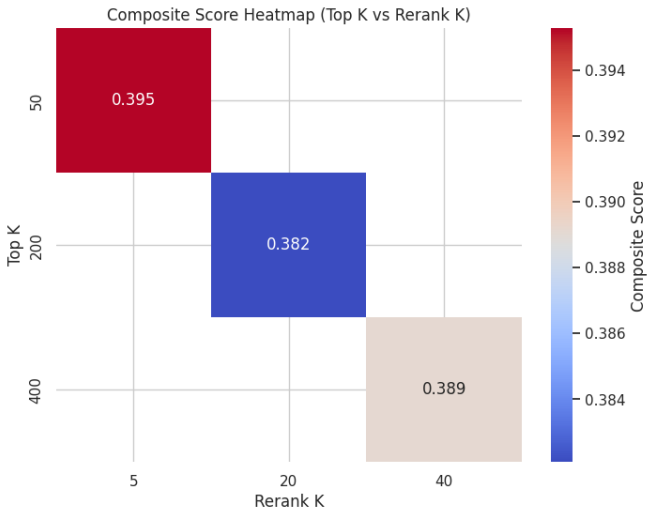


4.2 Retrieval Depth

Retrieval depth was varied by adjusting the number of retrieved passages:

Variant	Top K	Composite Score
retrieval_top50	50	0.395
retrieval_top400	400	0.389

The Top-K vs Rerank-K heatmap shows that increasing retrieval depth beyond moderate values produces diminishing returns. Retrieving too many passages may introduce irrelevant context, which can reduce generation quality and increase computational cost. Therefore, moderate retrieval depth combined with reranking generally provides a good balance between recall and precision.

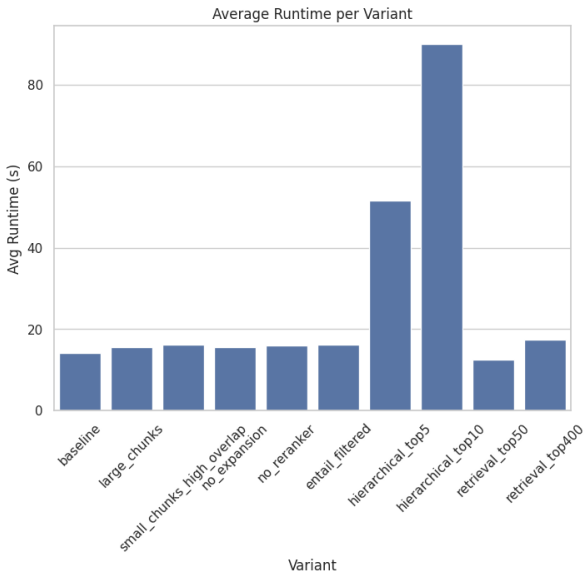


5 Runtime Efficiency

Average runtime for selected configurations is shown in the table below:

Variant	Runtime (s)
retrieval_top50	**12.48**
baseline	14.16
large_chunks	15.54
small_chunks_high_overlap	16.13
hierarchical_top10	**89.98**

As shown in Average Runtime per Variant figure, configurations that apply passage-level summarization before final generation (hierarchical variants) substantially increase runtime. These variants require multiple intermediate summarization steps before producing the final answer, which increases computational cost without yielding large improvements in evaluation metrics.

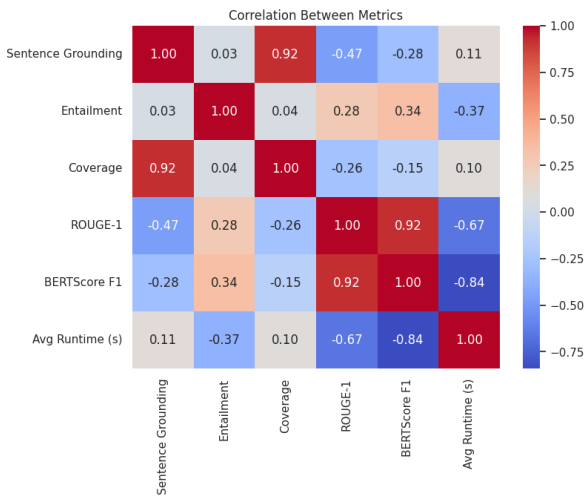


6 Metric Correlation Analysis

The relationships between evaluation metrics are visualized in Correlation Between Metrics. Strong heatmap. Two strong positive correlations appear:

- sentence grounding and coverage (≈0.92);
- ROUGE-1 and BERTScore (≈0.92).

These correlations indicate that semantically similar summaries tend to receive consistent scores across related metrics. A negative correlation is observed between runtime and semantic similarity metrics, suggesting that more computationally expensive configurations do not necessarily produce better summaries.

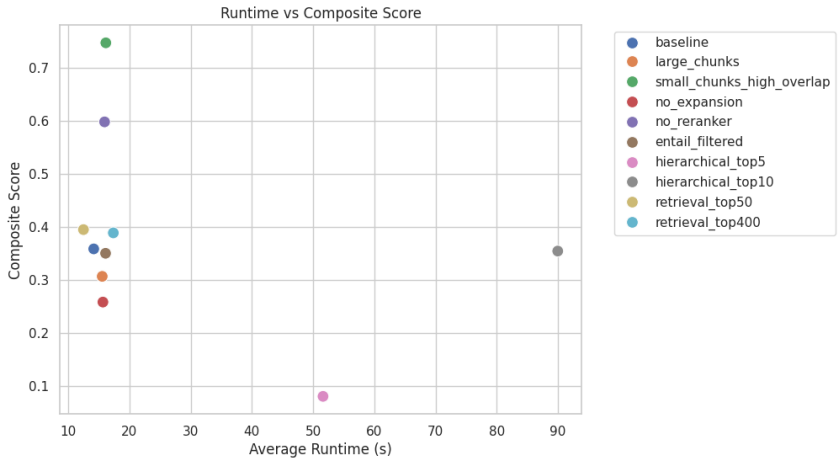


7 Runtime-Quality Trade-Off

The relationship between runtime and overall performance is illustrated in the Runtime vs Composite Score scatter plot.

Some configurations achieve relatively strong composite scores with moderate runtime. In contrast, configurations that perform additional summarization steps increase runtime significantly without substantial improvements in evaluation metrics.

This demonstrates that efficient retrieval configurations can provide strong performance without excessive computational cost.



8 Discussion

Several conclusions emerge from the conducted experiments.

First, retrieval configuration strongly influences RAG performance. Chunk size, overlap, and retrieval depth significantly affect grounding and coverage.

Second, smaller overlapping chunks provide more reliable retrieval evidence, improving alignment between generated sentences and retrieved passages.

Third, reranking can help refine retrieved passages, although its effect on reference-based metrics such as ROUGE is sometimes limited.

Finally, increasing generation complexity does not necessarily improve performance. Configurations that summarize many passages before generation dramatically increase runtime while providing limited gains in summary quality.

Overall, the results highlight that retrieval quality and document segmentation strategies are critical components of effective RAG systems.

9 Conclusion

This study evaluated multiple retrieval-augmented generation configurations for summarizing scientific papers. The experiments demonstrate that retrieval pipeline design strongly influences summary grounding, semantic similarity, and runtime efficiency.

The best performing configuration uses small overlapping chunks (100 words with 75-word overlap) combined with dense retrieval and reranking, achieving the highest composite score among all tested variants. Future work could explore: adaptive chunking strategies, improved retrieval filtering methods, multi-stage retrieval pipelines, and retrieval-aware training of language models.

Such approaches may further improve both summary quality and factual reliability in large-scale document understanding systems.